Lab Programs

SET 6- Product Sales Analysis

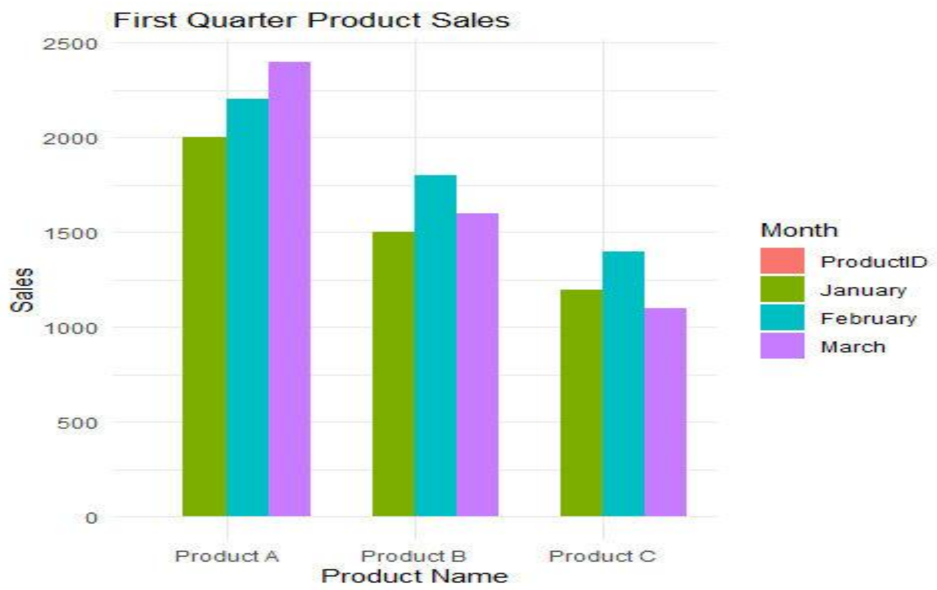
6A. Grouped Bar Chart

Program:

```
sales_data <- data.frame(
ProductID=c(1,2,3),
ProductName=c("Product A","Product B","Product C"),
January=c(2000,1500,1200),
February=c(2200,1800,1400),
March=c(2400,1600,1100)
)
sales_long <- melt(sales_data,
id.vars="ProductName",
variable.name="Month",
value.name="Sales")
```

```
ggplot(sales_long,
aes(x=ProductName,
y=Sales,
fill=Month))+
geom_bar(stat="identity",position="dodge")+
labs(title="First Quarter Product Sales",
x="Product Name",
y="Sales")+
theme_minimal()
```

Output:



6B. Stacked Area Chart

Program:

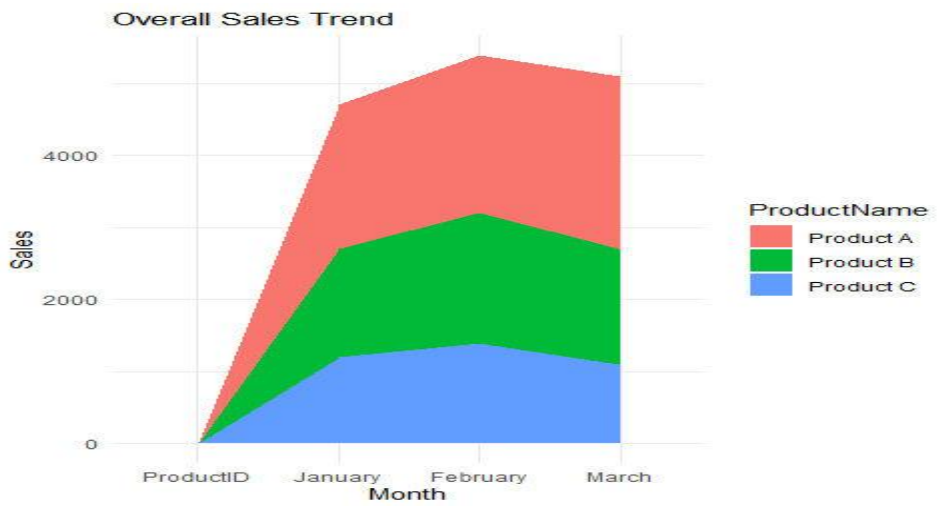
```
ggplot(sales_long,
aes(x=Month,
y=Sales,
fill=ProductName,
group=ProductName))+
geom_area()+
labs(title="Overall Sales Trend",
```

x="Month",

y="Sales")+

theme_minimal()

Output:



6C. Monthly Sales Table

Program:

sales_data

knitr::kable(

sales_data,

caption="Monthly Product Sales"

)

Output:

```
Table: Monthly Product Sales

| ProductID|ProductName | January| February| March|
|---------:|:-----------|-------:|--------:|-----:|
|         1|Product A   |    2000|     2200|  2400|
|         2|Product B   |    1500|     1800|  1600|
|         3|Product C   |    1200|     1400|  1100|
>
```

6D. Interactive Dashboard

Program:

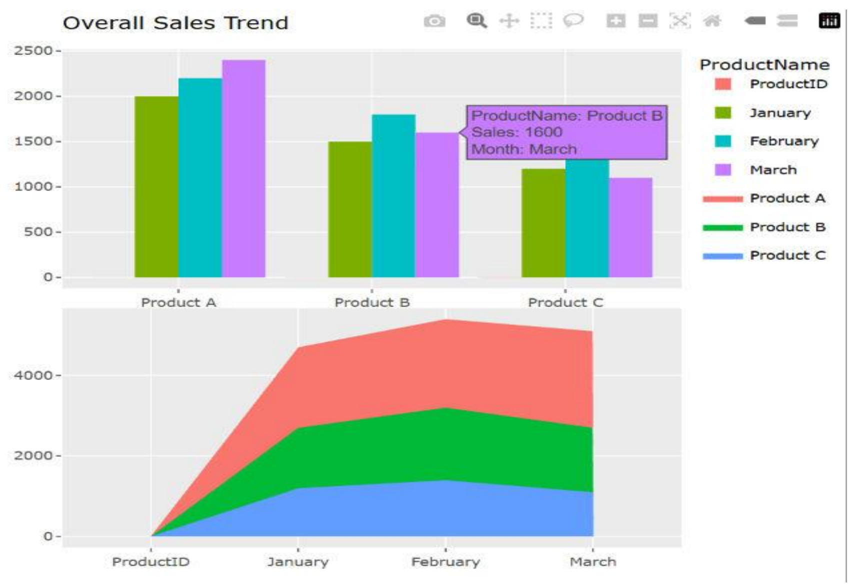
bar_plot <- ggplotly(

ggplot(sales_long,

aes(x=ProductName,

```
y=Sales,
fill=Month))+
geom_bar(stat="identity",position="dodge")+
labs(title="Quarterly Product Sales")
)
area_plot <- ggplotly(
ggplot(sales_long,
aes(x=Month,
y=Sales,
fill=ProductName,
group=ProductName))+
geom_area()+
labs(title="Overall Sales Trend")
)
dashboard <- subplot(bar_plot, area_plot, nrows=2) dashboard
```

Output:



SET 7- Customer Demographics Analysis

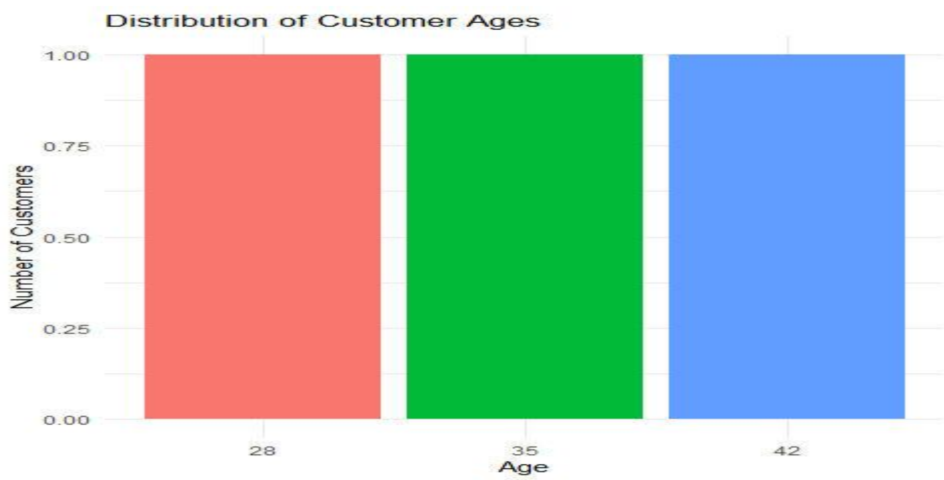
7A. Bar Chart

Program:

```
customer_data <- data.frame(
```

```
CustomerID=c(1,2,3),
Age=c(28,35,42),
Gender=c("Female","Male","Female"),
Income=c(50000,60000,75000))
ggplot(customer_data,
aes(x=factor(Age),
fill=factor(Age)))+
geom_bar()+
labs(title="Distribution of Customer Ages",
x="Age",
y="Number of Customers")+
theme_minimal()+
theme(legend.position="none")
```

Output:

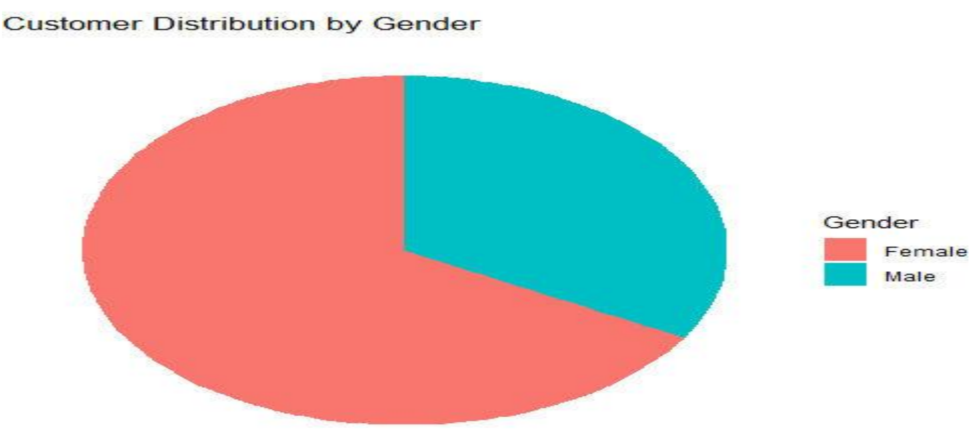


7B. Pie Chart

Program:

```
gender_data <- data.frame(table(customer_data$Gender))
colnames(gender_data) <- c("Gender","Count")
ggplot(gender_data,
aes(x="",y=Count,fill=Gender))+
geom_bar(stat="identity",width=1)+
coord_polar("y")+
labs(title="Customer Distribution by Gender")+
theme_void()
```

Output:



7C. Table

Program:

```
kable(
customer_data,
caption="Customer Demographics Information")
```

Output:

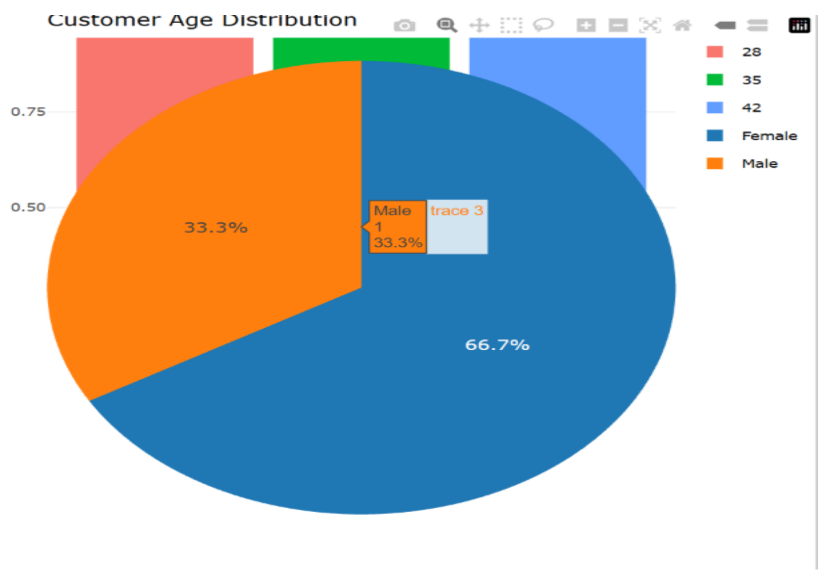
```
Table: Customer Demographics Information

| CustomerID| Age|Gender | Income|
|----------:|---:|:------|------:|
|          1|  28|Female |  50000|
|          2|  35|Male   |  60000|
|          3|  42|Female |  75000|
>
```

7D. Combined Interactive Dashboard

Program:

```
library(plotly)
bar_plot <- ggplotly(
 ggplot(customer_data,
     aes(x=factor(Age), fill=factor(Age))) +
  geom_bar() +
  labs(title="Customer Age Distribution") +
  theme_minimal() +
  theme(legend.position="none"))
gender_data <- data.frame(
 Gender=c("Female","Male"),
 Count=c(2,1))
pie_plot <- plot_ly(
 gender_data,
 labels=~Gender,
 values=~Count,
 type="pie")
dashboard <- subplot(bar_plot, pie_plot, nrows=2)
dashboard
```

Output:



SET 8- Employee Performance Analysis

8A. Line Chart

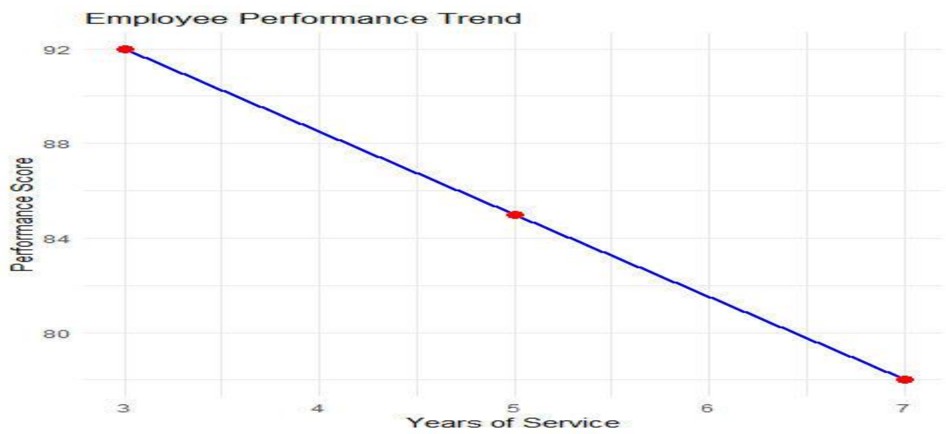
Step 1: Install Packages (Run Only Once)

```
install.packages("ggplot2")
install.packages("plotly")
install.packages("knitr")
```

Program:

```
employee_data <- data.frame(
EmployeeID=c(1,2,3),
Department=c("Sales","HR","Marketing"),
YearsOfService=c(5,3,7),
PerformanceScore=c(85,92,78))
ggplot(employee_data,
aes(x=YearsOfService,
y=PerformanceScore,
group=1))+
geom_line(color="blue",size=1)+
geom_point(color="red",size=3)+
labs(title="Employee Performance Trend",
x="Years of Service",
y="Performance Score")+
theme_minimal()
```

Output:

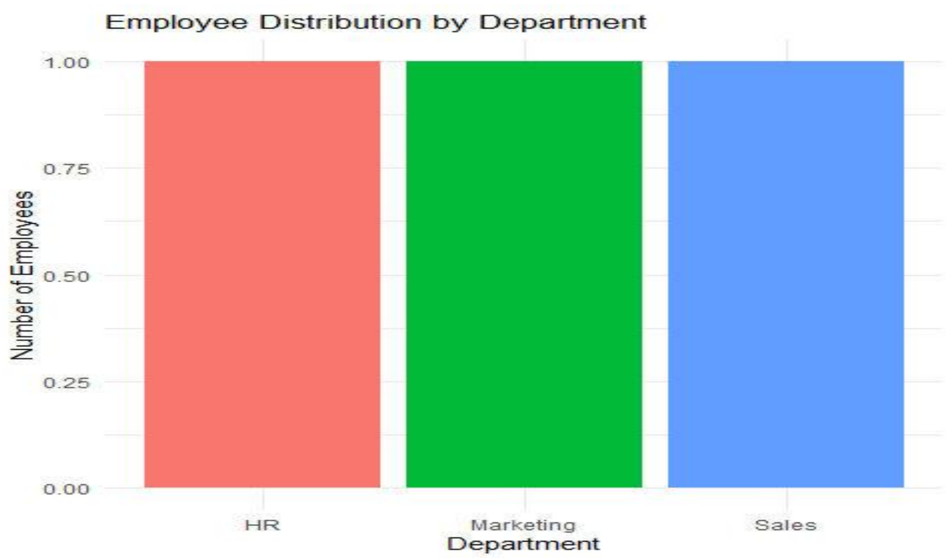


8B. Bar Chart

Program:

```
ggplot(employee_data,
aes(x=Department,
fill=Department))+
geom_bar()+
labs(title="Employee Distribution by Department",
x="Department",
y="Number of Employees")+
theme_minimal()+
theme(legend.position="none")
```

Output:



8C. Table

Program:

```
kable(
employee_data,
caption="Employee Performance Data")
```

Output:

```
Table: Employee Performance Data

| EmployeeID|Department | YearsOfService| PerformanceScore|
|----------:|:----------|--------------:|----------------:|
|          1|Sales      |              5|               85|
|          2|HR         |              3|               92|
|          3|Marketing  |              7|               78|
> |
```

8D. Interactive Dashboard

Program:

```
library(plotly)
line_plot <- ggplotly(
ggplot(employee_data,
aes(x=YearsOfService,
y=PerformanceScore, group=1))+
geom_line(color="blue")+
geom_point(color="red",size=3)+
labs(title="Employee Performance Trend"))
bar_plot <- ggplotly(
ggplot(employee_data,
aes(x=Department,
fill=Department))+
geom_bar()+
labs(title="Employee Distribution")+
theme(legend.position="none")) dashboard <- subplot(
line_plot,bar_plot, nrows=1,
shareX=FALSE,
shareY=FALSE)
dashboard
```

Output:

Set 9- Online_Learning_Activity.csv

9A. Import Dataset, Histogram & Boxplot

Program:

data <- read.csv(file.choose())

Histogram:

ggplot(data,aes(x=Quiz_Score))+

geom_histogram(binwidth=5,fill="skyblue",color="black")+

labs(title="Distribution of Quiz Scores",

x="Quiz Score", y="Number of Students")+

theme_minimal()

Boxplot:

ggplot(data,aes(x=Course,

y=Quiz_Score, fill=Course))+

geom_boxplot()+

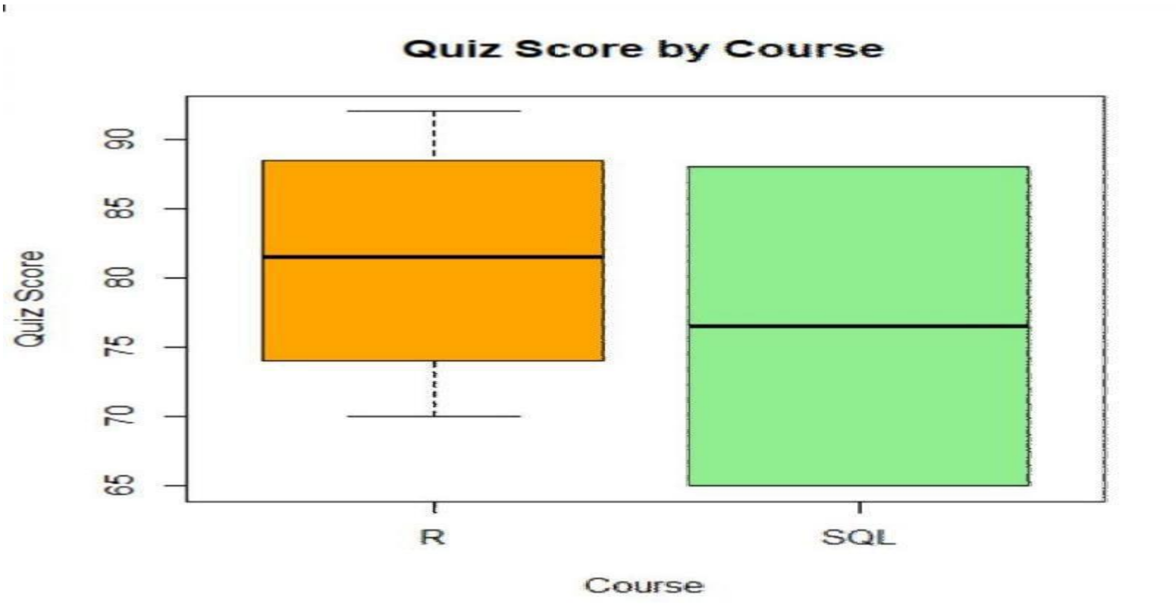
labs(title="Quiz Score by Course",

x="Course",

y="Quiz Score")+

theme_minimal()

Output:

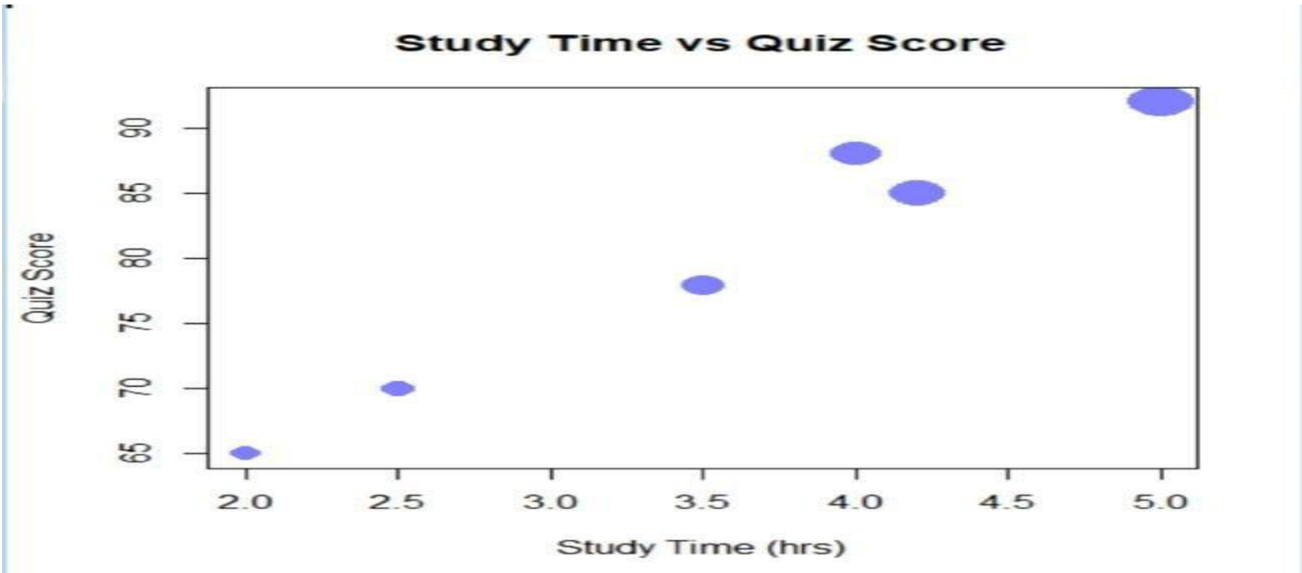


9B. Scatter Plot with Bubble Size

Program:

```
ggplot(data,
aes(x=Study_Time,
y=Quiz_Score,
size=Videos_Watched,
color=Course))+
geom_point(alpha=0.6)+
labs(title="Study Time vs Quiz Score",
x="Study Time (hrs)",
y="Quiz Score",
size="Videos Watched")+
theme_minimal()
```

Output:



9C. Date Conversion, Monthly Average & Line Chart

Program:

Convert Date

```
data$Login_Date <- as.Date(data$Login_Date)
```

Extract Month

```
data$Month <- format(data$Login_Date,"%Y-%m")
```

Average Quiz Score Per Month

```
monthly <- data %>%
group_by(Month) %>%
summarise(Avg_Quiz=mean(Quiz_Score))
```

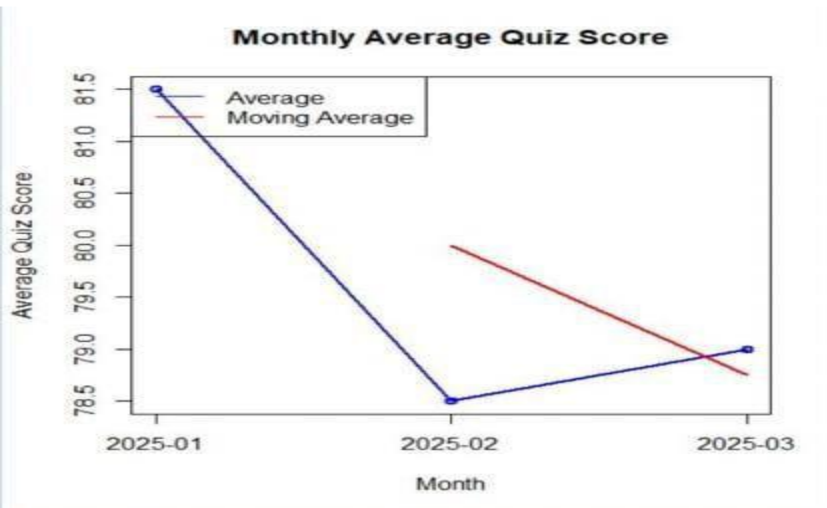
Moving Average

```
monthly$MovingAvg <- rollmean(
monthly$Avg_Quiz,
k=2,fill=NA,
align="right")
```

Line Chart

```
ggplot(monthly,aes(x=Month,y=Avg_Quiz,group=1))+
geom_line(color="blue",size=1)+
geom_point(size=3,color="red")+
geom_line(aes(y=MovingAvg),
color="green",
linetype="dashed",
size=1)+
labs(title="Average Monthly Quiz Score",
x="Month",
y="Average Quiz Score")+
theme_minimal()
```

Output:



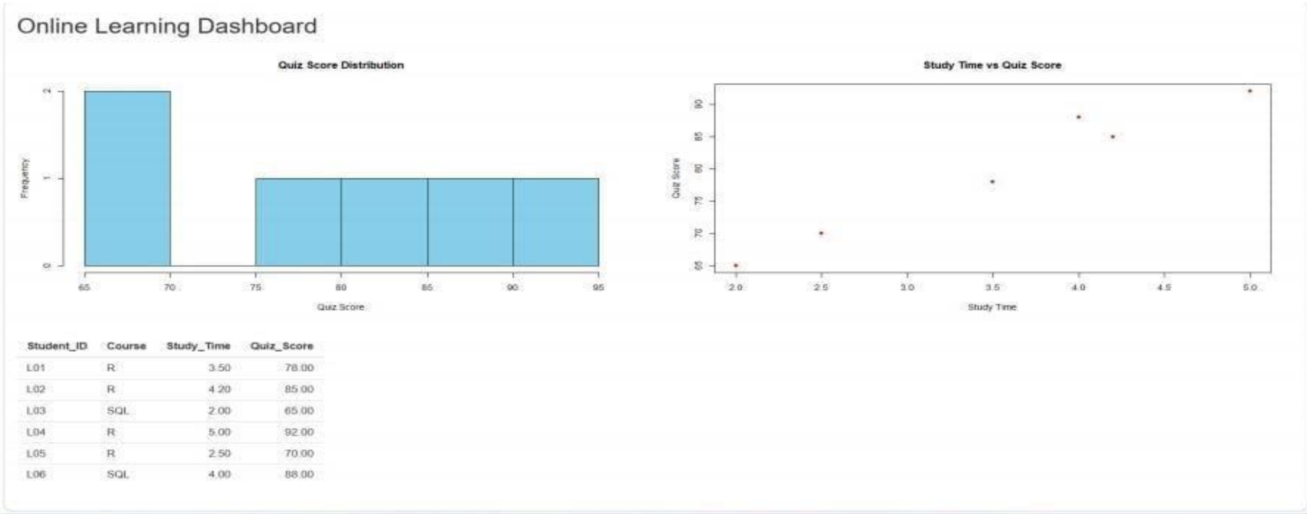
9D. Interactive Dashboard

Program:

```
library(shiny)
data <- data.frame(
Student_ID=c("L01","L02","L03","L04","L05","L06"),
Gender=c("Male","Female","Male","Female","Male","Female"),
Course=c("R","R","SQL","R","R","SQL"),
Study_Time=c(3.5,4.2,2.0,5.0,2.5,4.0),
Quiz_Score=c(78,85,65,92,70,88)
)
ui <- fluidPage(
titlePanel("Online Learning Dashboard"),
sidebarLayout(
sidebarPanel(
selectInput("gender",
"Gender",
choices=c("All",unique(data$Gender))),
selectInput("course",
"Course",
choices=c("All",unique(data$Course))) ),
mainPanel(
fluidRow(
column(6,
plotOutput("bar")),
column(6,
plotOutput("scatter"))
) ) )
)
server <- function(input,output){
output$bar <- renderPlot({
```

```
d <- data
if(input$gender!="All")
d <- subset(d,Gender==input$gender)
if(input$course!="All")
d <- subset(d,Course==input$course)
avg <- tapply(d$Quiz_Score,d$Course,mean)
barplot(avg,
col="skyblue",
main="Average Quiz Score by Course",
ylab="Quiz Score")
})
output$scatter <- renderPlot({
d <- data
if(input$gender!="All")
d <- subset(d,Gender==input$gender)
if(input$course!="All")
d <- subset(d,Course==input$course)
plot(d$Study_Time,
d$Quiz_Score,
pch=16,
col="red",
xlab="Study Time",
ylab="Quiz Score",
main="Study Time vs Quiz Score")
})
}
shinyApp(ui,server)
```

Output:



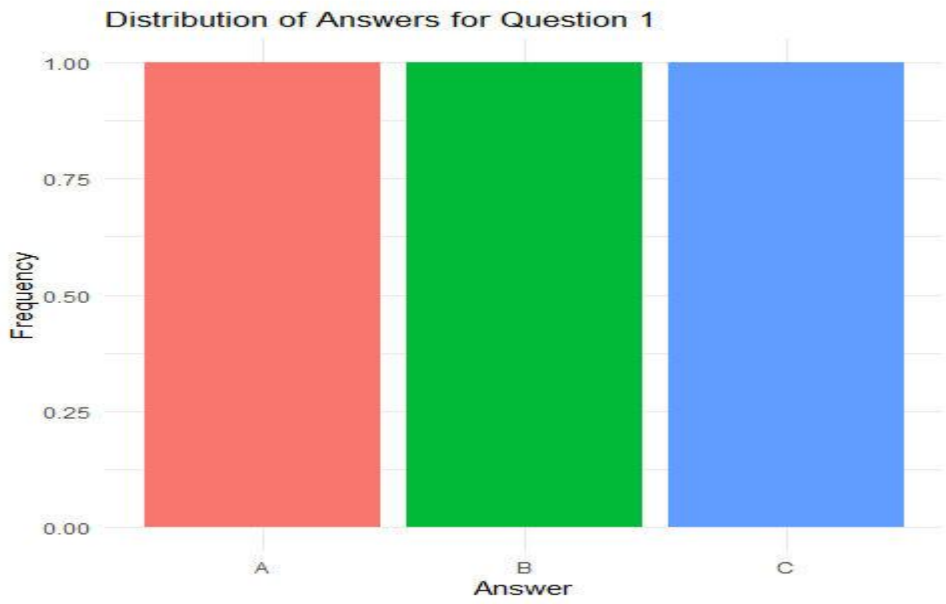
Set 10: Survey Responses Analysis

10A. Grouped Bar Chart

Program:

```
survey_data <- data.frame(
SurveyID=c(1,2,3),
Question1=c("A","B","C"),
Question2=c("B","A","A"),
Question3=c("C","D","B"))
q1_count <- as.data.frame(table(survey_data$Question1))
colnames(q1_count) <- c("Answer","Count")
ggplot(q1_count,
aes(x=Answer,
y=Count,
fill=Answer))+
geom_bar(stat="identity")+
labs(title="Distribution of Answers for Question 1",
x="Answer",
y="Frequency")+
theme_minimal()+
theme(legend.position="none")
```

Output:

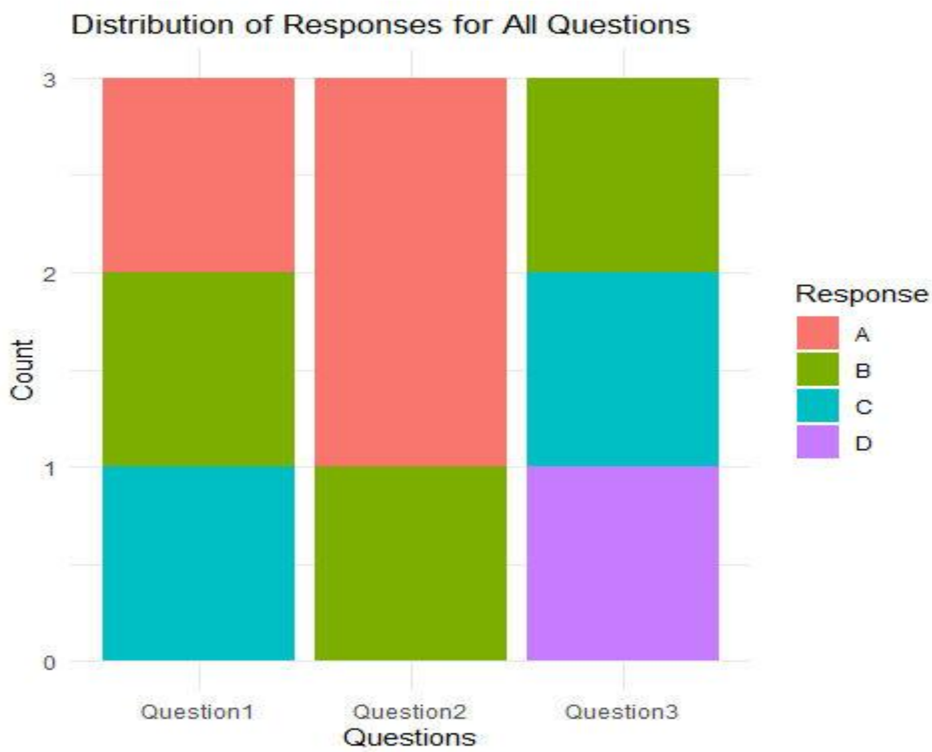


10 B. Stacked Bar Chart

Program:

```
survey_long <- melt(
survey_data,
id.vars="SurveyID",
variable.name="Question",
value.name="Response")
ggplot(survey_long,
aes(x=Question,
fill=Response))+
geom_bar()+
labs(title="Distribution of Responses for All Questions",
x="Questions",
y="Count",
fill="Response")+
theme_minimal()
```

Output:



10C. Table

Program:

kable(

survey_data,

caption="Survey Response Data"

)

Output:

```
Table: Survey Response Data

| SurveyID|Question1 |Question2 |Question3 |
|--------:|:---------|:---------|:---------|
|        1|A         |B         |C         |
|        2|B         |A         |D         |
|        3|C         |A         |B         |
> |
```

10D. Interactive Dashboard (R)

Program:

bar_plot <- ggplotly(

ggplot(q1_count,

```
aes(x=Answer,
y=Count,
fill=Answer))+
geom_bar(stat="identity")+
labs(title="Question 1 Responses")+
theme_minimal()+
theme(legend.position="none"))
stacked_plot <- ggplotly(
ggplot(survey_long,
aes(x=Question,
fill=Response))+
geom_bar()+
labs(title="Responses for All Questions")+
theme_minimal())
dashboard <- subplot(bar_plot,
stacked_plot, nrows=1,
shareX=FALSE,
shareY=FALSE)
dashboard
```

Output: